

Training Orchestral Conductors in Beating Time

Kathleen Griffin

Victoria University of Wellington
Wellington, New Zealand
k.g@outlook.co.nz

Neil A. Dodgson

Victoria University of Wellington
Wellington, New Zealand
neil.dodgson@vuw.ac.nz

ABSTRACT

Orchestral conducting involves a rich vocabulary of gestures and so training conductors is challenging. We discuss how virtual reality and gesture detection could be used to aid this process. We describe our pilot interface for training conductors in basic beat patterns, using gestural input and virtual reality output. We investigated both positional and acceleration-based detection of beats, concluding that the best way to detect beats reliably is to identify maxima in acceleration, that is, those moments in a beat that would appear as a flick to a human player. This practical evidence supports Wöllner's theory of how human players detect beats. We trialled our system with novices and experts, with a range of beating styles.

CCS CONCEPTS

• Human-centered computing → Gestural input.

KEYWORDS

gesture, music, conducting

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1 BACKGROUND

Conducting an orchestra is one of the most complex non-verbal communication mechanisms used by humans. Conducting is a domain in which gesture is vital to communication and in which a grammar has developed over more than a hundred years. The conductor is an expert whose gestures direct a hundred other experts. The conductor's job is to lead, inspire, and coordinate the ensemble. For some, it seems that a conductor plays the orchestra like a large instrument, but this is imprecise. Conducting is communication within context: each player interprets the gestures in accordance with the music on the stand in front of them, making the orchestra a truly corporate effort.

The conductor's gestural commands provide only high-level information to players. Gestures are principally needed at critical points: to indicate changes of tempo; to keep an orchestra in time with a soloist (the orchestra usually follows the soloist, not the

other way around); to provide confidence at tricky moments (for example, by cueing key entrances); to provide dynamic instruction (for example, indicating that a section is playing too loudly); and to provide information about interpretation (for example, legato vs staccato). This gestural language is most evident in rehearsal, where a conductor uses a wide range of gestures to modify playing style on the fly. In performance, the ensemble should be sufficiently confident and sufficiently rehearsed that the conductor provides reminders rather than adjustments.

Because conducting involves a gestural grammar developed over more than a hundred years, learning how to use conducting gestures is challenging and it takes several months for a novice to attain a secure technique [McElheran 1989]. The challenge is partly logistical: one can practise alone, which allows no feedback; work one-on-one with a tutor, which is expensive; or inflict oneself on an ensemble, which is even more expensive. Training is essential to become proficient in the gestural language of conducting [Rudolf 1950]. Our project aimed to determine whether we could provide some automated support in the earliest period of training: learning how to beat time effectively. We aimed to prototype something that would be as useful to a trainee conductor as a flight simulator is to a commercial pilot. This contrasts with conducting systems aimed at the general public which provide only a rough estimate of beat speed affecting playback of a recording [Schertenleib et al. 2004].

The conductor's art has traditionally been mystified in discussion and studied only through textbooks and practitioner testimony [Wöllner and Auhagen 2008]. While numerous books provide guidance and training in conducting [Farberman 1997; Grosbayne 1973; McElheran 1989; Roxburgh 2014; Rudolf 1950; Seaman 2013], the limitations of traditional approaches have recently been supplanted by lively new areas of technical, empirical research, particularly into the physical correlates of successful coordination [Luck and Nte 2008; Luck and Sloboda 2009; Luck and Toiviainen 2006; Wöllner et al. 2012]. Recent technological developments in motion tracking now allow us to capture the conductor's motion and patterns of acceleration in fine detail.

The most basic part of the conductor's art is beating time, but the technique goes well beyond this. A conductor must communicate with the ensemble, requiring a deep knowledge of the music, an ability to build rapport, clear verbal communication when not playing, and a clear, consistent gestural vocabulary when playing [McElheran 1989; Rudolf 1950]. We believe that recent advances in technology allow us to address training in the gestural vocabulary via an automated professional training system.

We built an experimental test bed comprising a Leap Motion hand tracker with visual output through either a large screen or a virtual reality headset, and aural feedback through speakers. We were guided by expert conductors and tutors of conducting. This work has correlates in elite sports training, where computer-mediated

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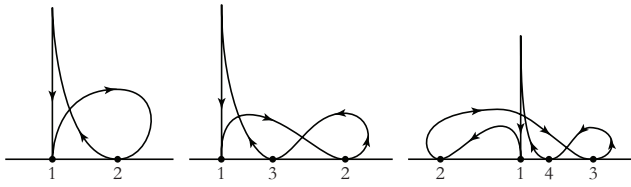


Figure 1: The three beat patterns we trialed (redrawn from McElerhen [1989]).

measurement is replacing traditional methods. Traditional methods rely on visual observation and simple timing data, which means that this “highly subjective and ‘expert’ driven approach is limited to the athletes’ and coaches’ interpretation of observed actions” [Fleming et al. 2010]. The same is true of conducting, but the challenges are greater, in that the training itself is extremely limited, with the trainee conductor then let loose to manage as best they can with only sporadic, anecdotal feedback to guide them.

Most previous work in gesture for music has been defined by what gestures the device can recognise, with most of that work considering how to use those gestures to create new ways to control music. ‘Gesture’ is a key term employed at NIME events (New Interfaces for Musical Expression), where its uses range from descriptions of human motion and system control, to metaphorical applications [Jensenius 2017]. Our work differs from that undertaken by the NIME community, in that we are aiming to codify gestures that are already in use in the established realm of conducting, and we explicitly understand that the gesture provides only partial control of the playback.

2 DETECTING BEATS THROUGH MAXIMA IN ACCELERATION

We ran a pilot project that addressed the simplest part of the conductor’s grammar: beating time with the right hand. We created a system that could detect beating of time in two, three or four (e.g., 2/2, 3/4, 4/4; Fig. 1). We based our beat identification on the theory developed by Wöllner et al. [2012], where beats are determined by maxima in the second derivative of position. That is, they are at the positions of greatest acceleration. Visually, these maxima appear to a human musician as a ‘flick’ in the beating pattern. We believe that musicians become attuned to detecting these flicks. Our pilot study indicated that these maxima appear to be both accurate (the beats are detected where you would expect them to be detected) and robust (they allow recognition of different beat patterns). This is in contrast to other work, which determines beats by minima in vertical position [Borchers 1997; Borchers et al. 2002; Segen et al. 2000], which we found to be less robust when we implemented it.

The advantage of using acceleration data, to search for those ‘flicks’, is that it is robust in the case of conductors whose beating styles have flicks that are to the left or right rather than vertically down. While all of our test conductors delivered a strong downbeat, we noted that their other beats could be in various directions, so testing for minima of vertical position would not be effective. Testing for acceleration allowed our system to detect reliably the beats in all five of our test conductors (two amateur, three professional)

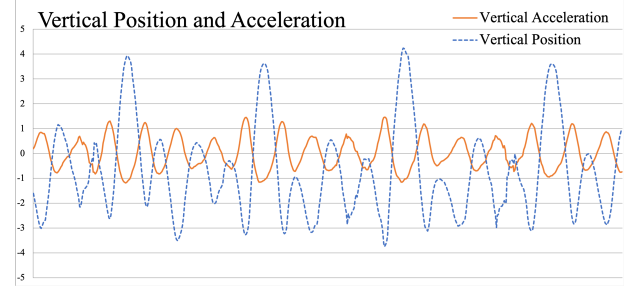


Figure 2: Four bars of beating (600 samples at 20ms intervals).

despite two of them have sufficiently idiosyncratic styles that they acknowledge that human musicians sometimes have problems following their beat.

However, the challenge with using a second derivative rather than position is that any noise in the positional data is enhanced by taking two derivatives. We therefore experimented with filtering on the data. Our final version used a 20ms sampling rate, with a 21-sample wide triangular filter on the second derivative. Fig. 2 shows the position and acceleration for an example of four bars of 4/4 beating by an amateur conductor who had a strong downward beating style on all four beats in a bar. For this conductor, detecting either minima of position or maxima of acceleration provides good beat detection. The difference between the two is that the acceleration version detects the downbeat on average 40ms before the position version, while the other three beats are detected to within ± 20 ms between the two versions. The acceleration version gives more consistent results from beat to beat (that is beats are more evenly spaced). This is hypothesised to be because the ‘flick’ on the strong downbeat occurs slightly before the conductor’s hand hits its minimum position.

Our system was developed to train McElerhen’s [1989] beat patterns. As noted above, it was equally able to work off the very different beating styles of our test conductors who were certainly not using McElerhen’s patterns but who are still putting ‘flicks’ into their beating to indicate the location of the beat. This makes sense: all human players will be looking for the same thing regardless of the conductor and, while players do adapt to a conductor’s style, it helps if the basic indicators are similar across all conductors.

Aural feedback is vital for the trainee. For our pilot project, we played back nursery rhymes, chosen because they have a steady rhythm with one note per beat. Our system played back the music in time with the beating input, allowing the user to hear whether they were giving steady input and to experiment with accelerando and decelerando.

An orchestra does not necessarily play at the exact position at which the conductors indicate the beat. Many conductors prefer that the orchestra play slightly after the beat so that the musicians react to the conductor’s indications. Our pilot system has a delay of about a quarter of a second, which felt right to our test conductors, and which allowed the signal processing to produce effective beat detection because it had information about the time after the beat as well as before. If we also want a mechanism where the delay is zero (e.g., one of our advisers has a preference for the orchestra to

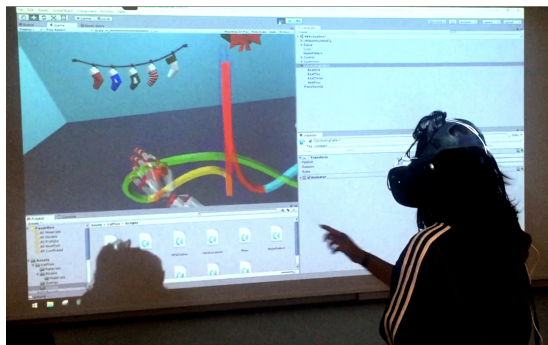


Figure 3: The VR system in use.

play exactly on his beat), we will need to develop algorithms that can predict beats rather than respond to them.

Any beat detection system must deal with imperfect humans providing the input, so not giving a clear beat on occasion. These lead to detection of extra beats and missing beats. We also had challenges with cleanly starting and stopping, and with handling upbeat (i.e., the beat before the music starts). While spotting missing or extra beats might appear straightforward, past research shows that it can be challenging even at a constant tempo [Borchers et al. 2002].

We used Unity, a VR headset, and a Leap Motion hand tracker to build two modes for interaction. A desktop version involves the motion detector on the desk in front of the computer's standard screen. The conductor gestures in the air above the motion detector with feedback on the screen in front of them. A virtual reality version allowed subjects to conduct in a VR environment with the motion detector mounted on the headset. Fig. 3 shows the VR system in operation with the development environment on the projected screen behind the user, so that others can observe what is happening in the VR space. The Leap Motion is attached to the VR headset allowing the subject to gesture freely in space. The subject's hand is echoed in the VR space by a robot hand. An outline of a McElerhan beating pattern is floating in the VR space for the user the practice against. This outline guide is an improvement over simply gesturing in empty space or, as Rudolf suggests, pasting a copy of the beating pattern onto the wall and waving your hand around in front of it. Both the VR and desktop versions proved to be useful environments and we note that there is uncertainty in the literature as to which is preferable for any given scenario [Demiralp et al. 2006; Prabhat et al. 2008].

Our system was used by two amateur and three professional conductors. The amateurs felt that the system improved their beating and they found themselves adapting their styles to the McElerhan [1989] beat patterns. The professionals were all impressed that the system was able to detect their beating, even though they each had a different beating style.

Fig. 4 shows an example of an amateur conductor, beating in 4, over 13 bars. Our system is able to give this kind of visual feedback in real time, with the path of the subject's hand being drawn in mid-air and remaining there for several bars. This allows a trainee to monitor their consistency. In this particular example, while the pattern is reminiscent of McElerhan's ideal, we see considerable

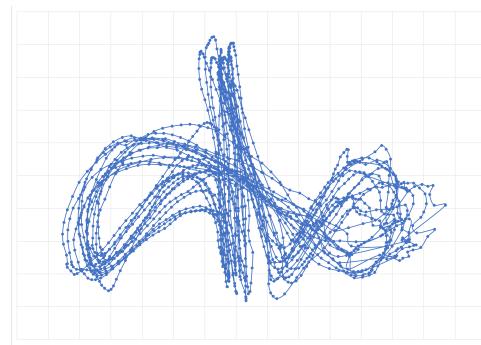


Figure 4: Example data captured from an amateur conductor.

variation from bar to bar that shows that a purely positional approach to beat-detection could be inappropriate.

3 FUTURE RESEARCH

This project provided valuable evidence that Wöllner's [2012] theory is correct about how humans detect beats, and showed that we had built a system that could provide useful training to amateurs and could handle variation in beating style. This could form the basis for a tutorial-style system for training basic beating.

However, there would be considerably more complex work required if we wished to develop a training environment that moved beyond this basic beating of time. In this section we outline the challenges that face further development.

The next steps would be to investigate the variations that can be made in the right hand's gestures to indicate articulation (legato, espressivo, staccato) and dynamic (forte, piano) [McElheran 1989]. Rudolf identifies non-espressivo, staccato, espressivo-legato and tenuto as sufficiently different to be worth a chapter each in his textbook [Rudolf 1950]. This requires more subtle gesture recognition and would need novel algorithms that are able to make fine distinctions reliably. Human musicians are clearly capable of making these distinctions, so it is not beyond the bounds of an automated system.

There are other parts of a conductor's art: the use of gaze direction, body positioning, and the appropriate use of the other hand to cue [Braem and Bräm 2000; Rudolf 1950]. These are vital aspects in conducting, where the conductor creates rapport with the individual musicians, providing direction specific to the part that the individual is playing. To address this challenge, we would need to go beyond simple feedback to create a simulated visualisation of an orchestra, to allow a trainee to get a good feel for where to look. In creating such a simulation, it would be useful to simulate the ways in which individual players actually respond to cues, e.g., one could consider a system where a player enters tentatively or incorrectly if not cued appropriately. We envisage a virtual reality system in which musicians are represented visually in their correct locations in the orchestra. This spatial arrangement is to train the conductor in where to look to provide appropriate cues. The visual representation can be simple (e.g., a tall rectangular box showing an image of the instrument) or a full dynamic simulation of human players. The former would be simpler and may be sufficient for

the purposes of training. The latter would be considerably more challenging and is reminiscent of a gamified version rather than something necessary for actual training.

Indeed, it has not escaped our notice that there is potential here for a gamified version that allows members of the general public to learn something about conducting. However, we are cautious about this: conducting is a gestural interaction mechanism that requires considerable training, so we know that any attempt to make it 'intuitive' for a novice will lead to considerable simplification and thereby restriction of what can be conveyed [Borchers et al. 2002; O'Hara et al. 2013].

A more ambitious extension would be playback of music where the played notes do not necessarily align with the beats (e.g., syncopation, off-beats), where there may be a change of tempo or a change of metre. This will require us to provide the system with not only the notes to play back but also information to allow the system to simulate an understanding of the score and put the conductor's gesture in context. For example, Rudolf [1950] provides an extensive collection of musical extracts chosen for training particular skills and these could be adopted into a training regime.

In terms of signalling other factors in the music, conductors tend to indicate changes and cues slightly ahead of time. Their job is to lead, so, for example, changes in volume are indicated a beat before they occur; cues might be two or three beats in advance. This means that any refinement of our system needs to respond with an appropriate delay.

Conductors are able and expected to provide appropriate cueing to different orchestral sections as necessary. This would require development of algorithms that respond to the subtle cues of gaze direction, body posture, and gestures with the left (non-beating) hand.

To achieve this, a fully-simulated system would require that each virtual musician has context in which they are playing and some machine intelligence to respond appropriately to cues (and lack of cues). Each virtual musician would be working from their particular (digitised) musical part, which provides both the notes to be played and information about dynamics, timing, etc. This is building on our earlier observation that the conductor provides overall direction to a set of musicians who each have individual instructions about what to play, when, and how. Rather than having perfect musicians, which would be straightforward to implement, it would be necessary to simulate the things that go wrong in real rehearsals, for example, with sections playing at the wrong volume, with players missing their entrances, and with players losing their place. We see the potential to develop a system that is similar to a flight simulator, allowing a tutor to put the trainee into situations that they will face in real life, and doing so in a controlled and repeatable manner [Hays et al. 1992].

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